KCC CODE CLUB · GAME MAKERS

Week 4

Decisions and Sensing

Week 4 of 8

Name _____

Today's big idea

The computer can ask a **yes-or-no question** about the world, and do different things depending on the answer.

```
if <touching the player?> then                    CONTROL
    ...do this...
```

```
else                                              CONTROL
    ...do that instead...
```

A question that can only be answered yes or no is called a **boolean**. Those blocks are **pointy hexagons**, and they only fit in pointy hexagon slots. Shape tells you what fits where.

Words to know

Word	What it means
condition	The yes/no question
boolean	An answer that can only be yes or no (true or false)
if / then	Do something only if the answer is yes
if / then / else	Do one thing if yes, a different thing if no
sensing	Blocks that let a sprite notice the world

> **Big one today:** an `if` block checks **once**, at the moment it runs. If you want it to keep checking, it has to be **inside a loop**.

Class activity — Bumper Sprite

```
when green flag clicked                           EVENTS
```

```
forever                                          CONTROL
    move 5 steps                                 MOTION
    if <touching [edge]?> then                   CONTROL
        turn 180 degrees                         MOTION
    wait 0.01 seconds                            CONTROL
```

Add a second sprite and make them bounce off each other too.

Your project — CATCHING!

Today your project becomes a real game you can actually play.

The new bit goes **inside the falling loop**:

```
repeat 20                                        CONTROL
    change y by -18                              MOTION
    wait 0.02 seconds                            CONTROL
    if <touching [Player]?> then
        start sound [Pop]                        SOUND
        go to x: (pick random -220 to 220) y: 170   MOTION
```

- [ ] Catching works
- [ ] A sound plays when you catch something
- [ ] Playtest it — then swap seats and play your neighbour's!
- [ ] **Save it**

Homework

1. Catch feedback. Make catching do something noticeable — a flash, a costume change, a different sound, a `say`.

2. Something you shouldn't catch. Add a bad falling object (a bomb, a coconut, a cat). For now it just says "Ouch!" when it hits you. Next week it'll cost you points.

Bonus challenges

1. **Traffic Light.** Add a red sprite. `forever` → `if <touching [Red]?> then say [STOP!] else say [GO!]`. Now drag your sprite in and out of the red one.
2. A sprite that follows the mouse, but only while space is held down.
3. Use `if` to stop your player leaving the stage — the thing we couldn't do in week 2!
4. Use `and` to combine two conditions: only catch it if you're touching it **and** pressing space.

If something breaks

- **Catching never works?** The `if` isn't inside the loop.
- **Works once, then never again?** The `if` is in the outer `forever` but not in the `repeat`.
- **Object falls straight through you?** You need the `go to` inside the `if` to send it back up.
- **Catching is really fussy?** Your object moves too far between checks. Make `change y by` smaller and the `repeat` number bigger.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.